

PROGRAM CONTROL	
RUN	Start the program
LIST	LIST 100 from a line
NEW	Erase program
CLEAR	Erase variables only
CONT	Resume after STOP / BREAK
STOP	Halt program
SAVE "name"	Save to cassette
LOAD "name"	Load (LOAD "" = next)
SLOW / FAST	Display on / compute-only
FAST blanks the screen while computing (like a ZX80); SLOW keeps the picture live.	

BASIC STATEMENTS	
LET	Assign a value
PRINT	; joins, , tabs
PRINT AT r,c	Row 0-21, col 0-31
PRINT TAB n	Tab to column n
INPUT	Prompt & read
IF..THEN	(no ELSE)
FOR..NEXT	FOR I=1 TO 9 STEP 2
GOTO GOSUB	Jump / call; RETURN
DIM	DIM A(20),A\$(5)
REM	Comment
RAND [n]	Seed RND
PAUSE n	Wait n frames (key ends it)
SCROLL	Scroll display up one line
POKE a,v	Write byte

FUNCTIONS	
ABS SGN INT	magnitude/sign/floor
SQR	square root
SIN COS TAN	radians
ASN ACS ATN	inverse trig
LN EXP PI	log / e^x / 3.14159
RND	random 0-1
PEEK(a)	read a byte
LEN CODE VAL	string len / 1st code / evaluate
STR\$ CHR\$	number↔char
INKEY\$	Scan keyboard (no wait)
USR addr	Call machine code
NOT AND OR	logical

GRAPHICS & DISPLAY		64×44
PLOT x,y	Light block: x 0-63, y 0-43	
UNPLOT x,y	Clear a block	
SCROLL	Move display up a line	
PRINT AT r,c	Text at row/col (24×32)	
CLS	Clear the screen	
Graphics are 2×2 block characters — an effective 64×44 grid. No colour, no sound.		

CURSOR / ENTRY MODES		keyword-per-key
K Keyword	— a key types its whole BASIC word	
L Letter	— normal typing	
F Function	— SHIFT+NEWLINE, one key (SIN, RND...)	
G Graphics	— SHIFT+9 toggles block chars	
The flashing cursor letter shows the mode. At a line start you're in K; numbers stay literal there.		

EDITING & KEYS		SHIFT +
EDIT	SHIFT + 1	— copy line to edit
RUBOUT	SHIFT + 0	— delete a character
Cursor	←→↵↻	SHIFT + 5 / 6 / 7 / 8
GRAPHICS	SHIFT + 9	— graphics mode
FUNCTION	SHIFT + NEWLINE	— function mode
BREAK	SPACE	while a program runs

MEMORY & POKES	
16384 (\$4000)	Start of system variables
D-FILE	The display (screen) file in RAM
16388/16389	RAMTOP (top of BASIC RAM)
PEEK / POKE	Read / write a byte
USR addr	Run machine code at addr
On a 1K machine the display file collapses to save RAM, so it grows as you print.	

REPORT CODES		m / line
0	OK / completed	
4	Out of memory	
5	Screen full	
6	Arithmetic overflow	
9	STOP executed	
D	BREAK pressed	
After a run the screen shows code/line at the bottom.		

QUICK-FACTS	
ZX80 (1980)	Integer BASIC · 4K ROM · 1K RAM
ZX81 (1981)	8K ROM · floating point · 1K RAM
TS1000	US ZX81 with 2K RAM
TS1500	ZX81 inside a Spectrum-style case, 16K
CPU / Video	Z80A ~3.25 MHz · mono RF · 24×32

ZX80 VS ZX81	
ZX80	Integer only; no SIN/COS, FAST/ SLOW, PLOT, SCROLL, PAUSE
ZX81 adds	Floating point + all of the above
Removed	TL\$ (use slicing / T0 instead)
The 8K ROM also upgrades a ZX80 (with a keypad overlay).	